🌸 CLASS 10 SCIENCE: LIFE PROCESSES

TOPPER NOTES

🌿 Biology Chapter 6 | 🍁 Nutrition & Respiration | 🌻 Hinglish Edition

1. What are Life Processes? 🦋

Definition: The basic maintenance functions performed by living organisms to prevent damage and breakdown are called **Life Processes**.

Hinglish Concept: Woh saare essential processes jo kisi living organism ko zinda rakhne aur uski body ki repair/maintenance ke liye zaroori hote hain, unhe Life Processes kehte hain. (Jaise: Nutrition, Respiration, Transportation, Excretion).

2. Nutrition (Poshann) 🍎

Definition: The process of intake of nutrients (food) and its utilization by the body for energy, growth, and repair.

Nutrition ke do main types hote hain:

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      NUTRITION / \ Autotrophic Heterotrophic (Self-food plants) (Dependent on others) - Photosynthesis -
Holozoic (Humans) - Green Algae - Saprophytic (Fungi) - Parasitic (Cuscuta)
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3. Autotrophic Nutrition (Swaposhan) 🌱

Meaning: Organisms prepare their own food from simple inorganic substances like CO_2 and water using sunlight.

Example: Green plants and blue-green algae.

💡 **Formula / Chemical Equation of Photosynthesis:**

$6CO_2 + 12H_2O \longrightarrow (\text{Sunlight / Chlorophyll}) \longrightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$

Three Main Events of Photosynthesis:

- **Step 1:** Absorption of light energy by chlorophyll.
- **Step 2:** Conversion of light energy into chemical energy & splitting of water molecules into Hydrogen (H_2) and Oxygen (O_2).
- **Step 3:** Reduction of carbon dioxide (CO_2) into carbohydrates (Glucose).

🚀 **Exam Tip (Stomata):** Stomata tiny pores hote hain leaves par jo gaseous exchange aur transpiration (water loss) mein help karte hain. Guard cells inke opening aur closing ko control karte hain (Water andar jaane par guard cells swell up hote hain aur pore khul jata hai).

4. Heterotrophic Nutrition & Amoeba Digestion 🦠

Meaning: Organisms jo apna khana khud nahi banate, balki plants ya dusre animals par depend hote hain.

Nutrition in Amoeba (Single-celled organism):

Amoeba **Holozoic nutrition** show karta hai steps mein: Ingestion → Digestion → Absorption → Assimilation → Egestion.

- Yeh **pseudopodia** (false feet) ko extend karke food particle ko capture karta hai aur **food vacuole** banata hai jahan digestion hota hai.

5. Human Digestive System 🫀

Human alimentary canal ek long tube hai jo mouth se anus tak hoti hai.

- **Mouth (Saliva):** Salivary amylase starch ko simple sugar mein todta hai.
- **Stomach:** Gastric glands secrete HCl, Pepsin (protein digesting enzyme), and Mucus (stomach wall ko acid se bachata hai).
- **Small Intestine:** Complete digestion of carbohydrates, proteins, and fats. Bile juice (liver se) fats ko emulsify karta hai.
- **Villi:** Finger-like projections in small intestine jo surface area badhate hain absorption ke liye.

6. Respiration (Shwasan) 🫁

Definition: The process of releasing energy from food broken down inside the cells is called respiration.

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      Glucose (6-Carbon) | (In Cytoplasm) ▼ Pyruvate (3-Carbon) + Energy | ├─► 1. Absence of Oxygen
(Yeast) → Ethanol + CO2 + Energy (Anaerobic) ├─► 2. Lack of Oxygen (Muscle cells) → Lactic Acid +
Energy (Cramps!) └─► 3. Presence of Oxygen (Mitochondria) → CO2 + H2O + Energy (Aerobic - 38 ATP)
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⚡ **ATP (Adenosine Triphosphate):** Yeh cell ki energy currency hai. Jab energy chahiye hoti hai, ATP break hokar ADP + Phosphate + Energy deta hai.

7. Differences: Aerobic vs. Anaerobic Respiration ⚖️

Feature	Aerobic Respiration 🌬️	Anaerobic Respiration 🧬
Oxygen	Takes place in the presence of O_2.	Takes place in the absence of O_2.
End Products	CO_2 and Water (H_2O).	Ethanol / Lactic Acid and CO_2.
Energy Released	High amount of energy (38 ATP).	Low amount of energy (2 ATP).
Location	Cytoplasm and Mitochondria.	Cytoplasm only (Yeast, Bacteria, Muscle cells).

💡 **Real-life Example of Muscle Cramps:** Jab hum bhari exercise karte hain, toh muscles mein oxygen ki kami ho jati hai (lack of oxygen). Isse pyruvate convert ho jata hai **Lactic Acid** mein, jisse muscles mein painful cramps ho jate hain!

8. Human Respiratory System 🫁

Path of Air: Nostrils ➔ Nasal Cavity ➔ Pharynx ➔ Larynx ➔ Trachea (Windpipe with cartilage rings) ➔ Bronchi ➔ Bronchioles ➔ Alveoli (Air sacs).

Mechanism of Breathing (Inhalation vs. Exhalation):

- **Inhalation (Saans lena):** Ribs move upward and outward, diaphragm flattens and moves downward. Chest cavity badi hoti hai, pressure kam hota hai aur air lungs ke andar chali jati hai.
- **Exhalation (Saans chhodna):** Ribs move downward, diaphragm domes up. Chest cavity choti hoti hai, pressure badhta hai aur CO_2 bahar nikal jati hai.

⭐ **Alveoli (Surface Area):** Alveoli mein bohot saari blood capillaries hoti hain jo gaseous exchange (O_2 absorption and CO_2 release) ke liye large surface area provide karti hain.

9. Exam Short Tricks & Mnemonics 💡

🚀 **Topper's Quick Tricks:**

- **Photosynthesis Formula:** 6, 12, 6, 6 yaad rakho! ($6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$).
- **Respiration breakdown:** Cytoplasm mein pehle Pyruvate banega (3-Carbon molecule), phir fate decide hoga oxygen ki availability par.
- **Cramps reason:** Lactic acid accumulation in muscles due to lack of O_2.

Frequently Asked Board Questions 🎯

- **Q. Why is diffusion insufficient to meet oxygen requirements of multicellular organisms like humans?**
  Ans: Multicellular organisms mein saare cells direct contact mein nahi hote environment ke, isliye simple diffusion se oxygen har cell tak nahi pahunch sakti. Respiratory pigments (Hemoglobin) zaroori hote hain.

⚡ Quick Revision Checklist (Last-Minute Summary) ✅

- [ ] **Life Processes:** Maintenance functions for survival.
- [ ] **Autotrophic:** Photosynthesis using sunlight, CO_2, and H_2O.
- [ ] **Stomata:** Gas exchange & transpiration control via guard cells.
- [ ] **Human Digestion:** Saliva $\rightarrow$ Stomach (HCl & Pepsin) $\rightarrow$ Small Intestine (Villi & Bile).
- [ ] **Respiration:** Glucose breakdown to release ATP energy.
- [ ] **Aerobic vs Anaerobic:** 38 ATP (with O_2) vs 2 ATP (without O_2).
- [ ] **Muscle Cramps:** Lactic acid formation due to lack of O_2.
- [ ] **Alveoli:** Site of gaseous exchange in human lungs.

🎉 All the Best for Your Board Exams! Score 100/100! 🌟